

Analyst Guide

This guide explains how analysts can operate the Nifty 100 Financial Intelligence Platform, navigate the Streamlit dashboard, use the financial screener, inspect company and sector fundamentals, work with peer comparisons, access reports, and use the REST API.

Validated platform status

- 92 companies in the analytical universe.
- 8 Streamlit dashboard screens.
- 16 FastAPI endpoints under /api/v1.
- 94 automated tests passing in the latest regression run.
- 5 KMeans company clusters assigned across all 92 companies.
- 92 company tearsheets, 11 sector reports, and a portfolio summary PDF generated.
- D43 integration test passed with 10 out of 10 concurrent screener requests successful.

Purpose

Use this document as the day-to-day operating guide for analysts and reviewers. It focuses on the practical workflow: start the services, navigate the dashboard, screen the universe, review company context, compare peers, inspect sector and cash-flow behaviour, and open primary documents when deeper investigation is required.

Platform Architecture

The platform combines a SQLite financial warehouse, Python analytics modules, machine-learning outputs, a Streamlit analyst interface, automated PDF reporting, and a FastAPI service layer.

Core data flow

- Source Excel datasets are normalised and loaded into data/nifty100.db.
- The analytics layer computes financial ratios, CAGR metrics, cash-flow intelligence, rankings, peers, and clustering outputs.
- The Streamlit dashboard reads the established analytical data layer and presents interactive analyst views.
- FastAPI exposes company, screener, sector, peer, market-cap, report, document, portfolio-statistics, and health endpoints.
- The dashboard Screener page also connects to FastAPI to verify service health and retrieve live screener preview data.
- Report modules generate company tearsheets, sector reports, and the combined portfolio summary.

Important directories

- data/ - SQLite database and source datasets.
- src/analytics/ - KPI, cash-flow, peer, ranking, clustering and intelligence logic.
- src/dashboard/ - Streamlit application and eight dashboard pages.
- src/api/ - FastAPI application and routers.
- src/reports/ - automated report generation.
- tests/ - ETL, KPI, API and data-quality test suites.
- docs/ - OpenAPI specification, Postman collection and analyst documentation.

Starting the Platform

1. Activate the virtual environment

```
.\venv\Scripts\Activate.ps1
```

2. Start FastAPI

```
python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

Swagger documentation is available at <http://127.0.0.1:8000/docs>. The health endpoint is <http://127.0.0.1:8000/api/v1/health>.

3. Start Streamlit

```
streamlit run src/dashboard/app.py
```

Run Streamlit in a second terminal while the FastAPI process remains active.

4. Confirm health

- The API health endpoint should return status healthy.
- The companies endpoint should return 92 companies.
- The Financial Screener sidebar should show FastAPI connected when the API service is running.
- A 404 response at <http://127.0.0.1:8000/> is normal because the project does not define a root route. Use [/docs](http://127.0.0.1:8000/docs) or [/api/v1/health](http://127.0.0.1:8000/api/v1/health).

Stopping services

Use Ctrl+C in the terminal that owns each foreground process. If FastAPI was started as a background process, stop that process explicitly before starting another copy on port 8000.

Dashboard Navigation

The analyst interface contains eight primary screens. Use the Streamlit navigation control to move between them.

1. Home

Portfolio-level financial and sector overview. Use it for broad orientation before drilling into individual companies.

2. Company Profile

Individual company fundamentals, financial ratio history, profit-and-loss trends, and generated business observations.

3. Financial Screener

Filter and rank companies using quality, growth, valuation, leverage and cash-flow conditions.

4. Peer Comparison

Compare companies within official peer groups using relative metrics and percentile-style comparisons.

5. Trend Analysis

Inspect multi-year KPI trends for a selected company.

6. Sector Analysis

Compare company-level fundamentals with sector benchmarks and medians.

7. Capital Allocation

Review operating, investing and financing cash-flow behaviour and the resulting capital-allocation classification.

8. Annual Reports

Browse available annual-report records and document links for primary-source follow-up.

Home and Company Profile

Home screen

- Select the financial year from the sidebar.
- Review the portfolio-level company and sector overview.
- Use the screen as the starting point for broad portfolio exploration.

Company Profile screen

- Search for a Nifty 100 company using the available identifier or company name.
- Review company-level ratio history and financial trends.
- Inspect profit-and-loss history across the available periods.
- Review generated pros and cons when available.
- Use profile information together with peer and sector analysis rather than relying on one KPI.

Interpretation note

Ratios should be interpreted within the company sector and business model. Banks, NBFCs, insurers, capital-intensive businesses and asset-light companies can have very different balance-sheet structures. Relative context matters.

Data-quality note

Some historical source data has mixed scales or unusual values. The project preserves source data and applies documented robustness steps in downstream analytics rather than silently rewriting raw records without evidence.

Financial Screener

The Financial Screener reduces the company universe using financial quality, growth, valuation and leverage conditions. The dashboard uses the established screener engine, while FastAPI provides the live service integration used in D43.

Common screening dimensions

- Return on Equity (ROE) - profitability relative to shareholder capital.
- Debt-to-Equity - leverage and balance-sheet risk.
- Free Cash Flow - internally generated cash after investment activity.
- Revenue CAGR - long-term sales growth.
- PAT CAGR - long-term profit growth.
- P/E Ratio - valuation relative to earnings.
- Sector - restrict analysis to more comparable businesses.

Preset screens

The project includes six preset screening styles: Quality, Value, Growth, Dividend, Momentum and Debt Free. Analysts can also apply custom thresholds.

Recommended workflow

- Start with broad thresholds so the universe is not over-filtered.
- Review how many companies remain after each major filter.
- Use composite ranking as a comparison aid, not as a standalone decision rule.
- Open shortlisted companies in Company Profile, Peer Comparison and Trend Analysis before drawing conclusions.

Peer Comparison and Trends

Peer Comparison

- Select a peer group containing economically comparable companies.
- Review company metrics against peer-group averages and percentile positions.
- Use radar visualisation to compare multiple financial dimensions simultaneously.
- Avoid comparing unrelated business models simply because they have similar market capitalisation.

Trend Analysis

- Select a company and one or more financial KPIs.
- Review the available annual history and look for persistence rather than one-year spikes.
- Compare profitability trends with revenue growth, leverage and cash-flow behaviour.
- Investigate sudden changes using annual reports and document links.

Useful analyst questions

- Is profitability improving consistently?
- Is growth supported by operating cash flow?
- Is leverage rising faster than earnings?
- Is the company outperforming or underperforming its peer group?
- Does the trend agree with the current screener classification?

Sector and Capital Allocation

Sector Analysis

Use Sector Analysis to compare company fundamentals with sector-level benchmarks. Sector medians reduce the risk of making inappropriate cross-industry comparisons.

- Review company count and sector-level medians.
- Use sector context when interpreting ROE, P/E, debt and growth.
- Treat sector comparisons as context, not a guarantee of quality.

Capital Allocation

- Positive operating cash flow shows cash generated by normal operations.
- Negative investing cash flow often represents reinvestment or asset purchases.
- Financing cash flow helps identify borrowing, repayment and shareholder distributions.
- The platform classifies common patterns such as Reinvestor, Shareholder Returns, Distress Signal, Growth Funded by Debt, Cash Accumulator and other mixed cases.

Distress review

Cash-flow distress flags are screening signals. Analysts should inspect the underlying company statements and annual reports before treating a flag as a conclusion.

Annual Reports and REST API

Annual Reports

- Select a company to view available annual-report records.
- Use document links to move from quantitative signals to primary-source investigation.
- Annual-report availability can vary by company and financial year.

REST API

FastAPI exposes 16 versioned endpoints under /api/v1. Interactive documentation is available through Swagger at /docs.

- Company endpoints: /companies, /companies/{ticker}, /pl, /bs, /cashflow, /ratios, /documents, /tearsheet.
- Analytics endpoints: /screener, /sectors, /sectors/{sector}/companies, /peers/{group_name}, /companies/{ticker}/peers/compare, /market-cap/{ticker}, /portfolio/stats.
- Operations endpoint: /health.

API usage note

API responses are JSON except the tearsheet route, which returns a PDF stream. Invalid company identifiers return HTTP 404.

Reports and Analytical Outputs

Automated reports

- 92 company tearsheet PDFs in reports/tearsheets/.
- 11 sector-level PDF reports.
- Portfolio summary PDF covering the full available company universe.
- Peer-comparison workbook and radar-chart outputs.

Key analytical files

- output/cashflow_intelligence.xlsx - cash-flow quality, capital intensity and distress analysis.
- output/pros_cons_generated.csv - rule-based company strengths and concerns.
- output/cluster_labels.csv - five machine-learning cluster assignments.
- output/clustering/cluster_profiles.csv - aggregate cluster characteristics.
- docs/openapi.json - exported API specification.
- docs/postman_collection.json - API request collection.
- pytest_report.html - automated test report.
- perf_notes.md - D43 integration and performance results.

Using reports

Use generated reports for consistent review and communication, but verify material conclusions against underlying statements and company disclosures.

Quality Assurance and Performance

Automated testing

- 94 tests are currently passing.
- The latest full regression run completed in 4.03 seconds.
- The suite covers ETL, KPI calculations, API behaviour and data quality.
- The API exposes exactly 16 /api/v1 endpoints.
- Invalid company identifiers return HTTP 404 as expected.

D43 integration and load test

- API health returned HTTP 200 and status healthy.
- The dashboard API client retrieved 92 companies.
- The tested screener query returned 57 companies.
- 10 concurrent screener requests completed successfully with 0 failures.
- Average observed latency: 1466.63 ms.
- Maximum observed latency: 1707.58 ms.
- Total concurrent-test wall time: 1.72 seconds.

Regression discipline

Run the full pytest suite after code changes and before final delivery. A generated HTML test report provides pass/fail counts and tracebacks for failures.

Analyst Workflow and Troubleshooting

Suggested analytical workflow

- Begin with Home or Sector Analysis to understand the broader universe.
- Use the Financial Screener to create a manageable shortlist.
- Open shortlisted companies in Company Profile and Trend Analysis.
- Compare each company with its official peer group.
- Review cash-flow behaviour and capital-allocation patterns.
- Use annual reports to investigate material changes or unusual metrics.
- Use tearsheets and generated reports for consistent documentation.

Troubleshooting

- If FastAPI is unavailable, confirm a Uvicorn process is running on 127.0.0.1:8000.
- If port 8000 is already in use, do not start a second server; verify the existing `/api/v1/health` endpoint.
- If Streamlit imports fail, run commands from the project root with the virtual environment active.
- If dashboard data looks stale, clear Streamlit cache or restart the application.
- Run the full pytest suite before final delivery or after any code change.

Final note

The platform is an analytical decision-support system. Results should be interpreted together with business context, sector characteristics, data-quality notes and primary company disclosures.